

CALM Drone Camera Specification & Requirements

System Specification:

1) Gimbal

- Azimuth: $\pm 140^\circ$
- Elevation: $+20^\circ$ to -130°
- Slew rate: 0.1° to $90^\circ/\text{sec}$
- Acceleration: $60^\circ/\text{sec}^2$
- 18-bit encoder for position accuracy
- 2-axis stabilization with $300\mu\text{rad}$

2) EO Camera requirement

- CMOS sensor: Color, Global Shutter
- Resolution: 1280 x 800 pixels
- Pixel size: 2.8μ
- 2 FOV camera: 35° and 11°
- Digital zoom: Continuous digital zoom between 35° to 4° .
- EO angular resolution: $<0.16 \text{ mrad /pixel}$
- Video O/P: USB 3 in YUV format
- Frame rate: 30fps
- Target Identification range: 800m (For Tank size target)
- No. of Pixels for 6m target at 800m: 47 pixels
- No. of Pixels for 3.6m target at 800m: 17 pixels
- Video o/p in YUV format over USB with imaging time stamp in form of stream

3) IR Camera

- Resolution: 640 x 480 pixels
- HFOV: 17°
- Video O/P: USB 3.0
- Frame rate: 25 fps
- Target Identification range: 500m
- No. of Pixels: 25 pixels (For 6m tank size target) and 15 pixels (For 3.6m size target)

4) Output Interface:

- Single USB interface with type C connector for gimbal control, IR and EO video o/p

5) Control & Software Interface:

- Gimbal Communication command update rate: 100 Hz

- Gimbal and Camera control ICD implementation at the OBC of CALM for camera, and gimbal control.
- Software should be Mavlink compatible

6) Electrical Requirement:

- I/P voltage: 28V
- Power consumption: $\leq 28W$
- I/P Power: 1 High 1 Low (2 pin Molex Connector)

7) Mechanical Mounting Interface:

- Mechanical Mounting with $\phi 55mm$ PCD (as per step file)

8) Physical Characteristics

- Weight of Gimbal $< 500g$
- Dimensions: $\phi 75mm \times 120mm$ (L)
- Operating Temp: $-30^{\circ}C$ to $+55^{\circ}C$

9) Additional Information Regarding UAV

- UAV max speed: 47m/s & Altitude: 800m AGL
- Launcher: Pneumatic Launch, exit velocity of 25.7 m/s
- UAV angle of attack: ranges from 7° to -11° depending on the flight phase.
- Launching vibration tolerance: 15g

